

HENRY GROVES

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Education

Further Study, University of Newcastle upon Tyne

2023-2027

- M.Sc.+B.Sc. (Hons) Computer Science

A Levels, Pontefract New College, Pontefract

2021-2023

- **A Levels:** Mathematics (A*), Further Mathematics (A), Computer Science (A), Physics (A)
- **EPQ:** “To what extent will quantum computing replace high performance computing in computational biology?” (A)

GCSEs, Carleton High School, Pontefract, West Yorkshire

2016-2021

- GCSEs: 5 9's, 3 8's
- BTECs: D*2, D2

Work Experience

Northern Powergrid — *Returning Summer Internship*

- **Data Scientist (DSO)** June–September 2026
 - Designed a shared development framework for an expanding team of data scientists, built on Reproducible Analytical Pipeline (RAP) principles and integrated into CI/CD pipelines on Azure DevOps, with standardised logging and configuration management, resulting in streamlined onboarding and collaboration.
 - Migrated 2 legacy manual processes previously in PowerBI/Excel to RAP pipelines, reducing run time and manual effort from multiple days work to a few minutes.
- **Data Scientist (DSO)** May–September 2025
 - Built a graph-based simulation tool modelling high-voltage networks under normal and fault conditions, used by many teams for both network planning and predicting LCT rollout.
 - Replaced a top-down approach with bottom-up simulation from historical and real-time asset data, enabling finer-grained analysis.
 - Modelled grid evolution to 2050 under 4 future energy scenarios.
- **Data Scientist (DSO)** May–September 2024
 - Designed an end-to-end ML pipeline forecasting short-term load across 800 distribution substations from half-hourly meter data, with gap-filling via Akima interpolation and lag-based methods, used for network planning and procuring flex.
 - Clustered load profiles with DTW k-means and showed that one model per cluster generalised across its members, reducing the number of models from 800 to 6.
 - Designed and tuned both **TCN** and **BiTCN** neural-networks, for each cluster, with carefully selected **exogenous features** boosting model performance.
- **Data Analyst (System Forecasting)** May–August 2023
 - Built a statistical demand model for 4M customers on low-voltage networks using SQL and Python, informing short-term and long-term business planning.
 - Prepared data for regulatory submissions to tight deadlines alongside model development.

Projects

“Biggmarket” - Item swapping app promoting sustainability

Built a full-stack item-swapping web application designed to promote sustainable consumption. Built using Next.js, TypeScript, tRPC, Tailwind CSS and backed by a PostgreSQL with Prisma, the app uses a “swipe left-right” interface, with real-time user messaging, authentication via Google OAuth, and a comprehensive user review system.

Conference/Oral Presentations

Robertson, B-A*; Jackson, L.; Groves, H.; and Hance, J.R. *Emotional Modulation of Temporal Order Memory in Childhood: A Quantum-Informed Account*. At Quantum Information and Probability: from Foundations to Engineering (QIP26), 10 June 2026. Växjö, Sweden. *presenter

Key Skills

- Problem-Solving
- Attention to Detail
- Procedural Thinking
- Software Development
- Machine Learning and AI
- Data Exploration & Analysis

Programming Languages / Frameworks:

- Python 3
- Rust
- R
- numpy / pandas / scikit
- tensorflow / pytorch
- matplotlib / seaborn
- C/C++
- TypeScript
- React.JS / Next.JS
- Java

Additional Technologies:

- GNU/Linux
- LaTeX
- Git / GitHub
- SQL (PostgreSQL)
- MongoDB
- HTML/CSS
- tailwindcss
- Access / Excel

Languages:

- English (Native)
- German (B1)